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Research Statement

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Research Agenda

My research asks how farms, markets, and institutions can better account for natural capital, a set of inputs that production depends on but that prices rarely capture. I combine causal inference with careful measurement to estimate policy-relevant parameters and, where possible, connect reduced-form evidence to structural models of how natural capital accumulates and is used. Two questions organize the agenda: what private incentives do producers face to build natural capital such as soil, and what is the economic value of biodiversity and ecosystem services that production and well-being rely on? Both aim to produce credible evidence that helps governments and organizations allocate scarce resources more effectively.

Job Market Paper: Regenerative Agriculture and Farmer Profitability

Question and context. In *Regenerative Agriculture and Farmer Profitability: Modeling Adoption for Scalable Climate Mitigation*, I ask whether conservation practices that can mitigate climate change are also privately profitable for farmers. The answer matters because the mitigation potential of practices such as cover cropping and conservation tillage depends in part on whether farmers adopt them without subsidy. I study Midwestern corn, soybean, and wheat production, and treat soil organic carbon (SOC) as a form of natural capital that farm management builds over time.

Data and design. I combine farm financial records for 2000–2025 with a machine-learned county-year panel of predicted SOC. Using a doubly robust staggered difference-in-differences design, I estimate practice- and crop-specific effects on net returns, revenues, expenses, yields, insurance outcomes, and predicted SOC.

Findings. The results reveal a heterogeneous profit premium rather than a uniform one. Estimated net returns rise by $79.40 per acre for corn cover crops and by $22.50 per acre for soybean reduced tillage, while most other practice-crop comparisons vary considerably. I also recover the long-run build-up of SOC from conservation agriculture, and it corresponds well with the financial benefit generated from soil capital accumulation.

Dynamic model. To understand long-run soil capital dynamics, I develop a dynamic structural model linking management, production costs, and insurance to soil-stock accumulation and future returns. The model separates observed private returns from agronomic potential and clarifies the intertemporal incentives that matter for conservation policy, including when private profit alone is enough to sustain adoption and when it is not.

Contribution. The paper (i) grounds the profitability of conservation practices in farm financial records, by practice and crop, (ii) pairs machine-learned SOC with a modern staggered-adoption design to measure the soil-capital channel itself, and (iii) links reduced-form estimates to a structural model that connects short-run returns to long-run incentives and the scalability of climate mitigation.

Valuing Biodiversity and Ecosystem Services

Willingness to pay for biodiversity. In *Willingness to Pay for Biodiversity: Evidence from Citizen Science* (with Raahil Madhok, Matthew Braaksma, and Jovin Lasway), we use panel data on birder trips from the eBird platform in India and a mixed-logit travel-cost model to recover willingness to pay for species richness. To

strengthen identification, we exploit quasi-experimental migration shocks from Siberian bird movements into India as an instrument for local biodiversity variation. This design yields causal, spatially and temporally heterogeneous valuations that we map into welfare changes for local and broader populations.

Making nature visible in economic accounts. A second stream builds accounting frameworks that put ecosystem services into aggregate economic statistics. In *Global Gross Ecosystem Product (GEP)* (with NatCap TEEMs and the Chinese Academy of Sciences' RCEES), we construct comparable estimates of ecosystem-service value for 250 countries and territories in 2019 by pricing and aggregating 35 services while avoiding double counting. I apply the framework to specific services. In *Economic Contribution of Wild Pollinators: An Application of GEP* (with Subin Poudel and Lingling Liu; under review at *Ecosystem Services*), I combine pollination sufficiency, crop-level dependence ratios, yields, and producer prices to estimate pollination-dependent production for 65 crops and identify hotspots where pollinator losses would be most costly. *Ecosystem Services to Own Consumption Crop Production* (with Marta Sylla and Lijing Wang) extends the method to subsistence systems often omitted from national accounts, using a 99-country panel (2004–2021). A further application values daily recreation and nature-based tourism (with Matthew Braaksma, Spencer Wood, and Lifeng Ren). Across these papers, the common contribution is a reproducible strategy for integrating natural capital into policy-relevant economic aggregates.

Ongoing Work

The value of scenic quality. In *What a View! The Economic Value of Scenic Quality* (with Matthew Hockert), I extend my valuation work to landscape amenities. We assess the hedonic value of nature landscape visibility through home prices in discrete choice modeling framework. Our analysis reveals a large divide in value driven by scarcity of natural features. A one standard deviation increase in scenic quality raises home values by approximately 2 percent in rural communities and 10 percent in urban areas.

The pollinator–pesticide tradeoff. Modern agriculture pairs rising demand for pollinator-dependent crops with input choices that can reduce wild pollinator stocks. In *The Pollinator–Pesticide Tradeoff in Agricultural Production*, I study how farmers trade off purchased managed pollination, pesticide use, and reliance on local pollination services, to identify the market forces behind potentially inefficient input bundles. The project aims to quantify the shadow value of common-pool natural inputs and to inform policies that jointly address productivity and biodiversity conservation.